

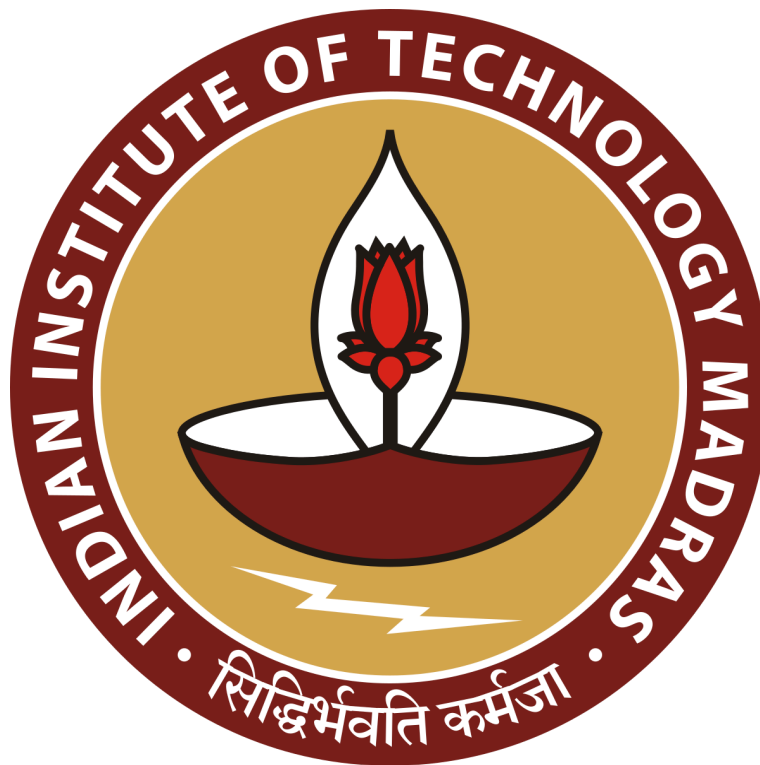
AI Agent for Academic Guidance

Project Report for Milestone 1 of Software Engineering

Submitted by

Team Number: Team 6

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Table of Contents

1. Types of Users.....	2
<i>1.1 Primary Users.....</i>	<i>2</i>
1. Students.....	2
2. Professors.....	2
3. Instructors.....	2
4. Teaching Assistants (TAs).....	3
<i>1.2 Secondary Users.....</i>	<i>3</i>
5. Admins.....	3
6. Degree Managers.....	3
<i>1.3 Tertiary Users.....</i>	<i>3</i>
7. Developers.....	3
8. Management.....	3
2. User Stories.....	4
2.1 Students.....	4
2.2 Professors.....	5
2.3 Instructors.....	6
2.4 Teaching Assistants (TAs).....	6
2.5 Admins.....	7
2.6 Degree Managers.....	7
2.7 Developers.....	7
2.8 Management.....	8
3. Team Details.....	8

1. Types of Users

1.1 Primary Users

1. Students	• View course materials, assignments, and announcements. • Submit assignments and track deadlines. • Interact with the AI assistant chatbot for learning support. • Communicate with instructors and teaching assistants for doubts.
2. Professors	• Manage course content, lectures, and assignments. • Monitor student performances and activities in the course. • Interact with the AI assistant chatbot to streamline tasks, personalize learning, and enhance teaching and collaboration. • Interact with Instructors, TAs, and Management.
3. Instructors	• Assist professors in managing course content. • Collaborate with teaching assistants for student support. • Interact with the AI assistant chatbot to take care of routine tasks, track student progress, and provide additional support for challenging concepts.

4. Teaching Assistants (TAs)	• Act as guides for students and support their learning process. • Communicate with instructors and students to resolve issues.
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1.2 Secondary Users

5. Admins	• Manage user roles and permissions. • Perform CRUD operations on platform resources.
6. Degree Managers	• Monitor overall platform activity and performance. • Perform CRUD operations on courses, groups, and spaces.

1.3 Tertiary Users

7. Developers	• Manage the codebase and implement new features. • Resolve bugs and ensure stability.
8. Management	• Oversee platform performance and ensure scalability. • Make strategic decisions for platform enhancements.

2. User Stories

2.1 Students

- As a student, I want to access course lectures and tutorials, so that I can learn at my own pace and ensure consistent progress in my learning journey.
- As a student, I want to access supplementary content, including slides, transcripts, notes, and reference books, so that I can refine my understanding on specific topics.
- As a student, I want to access and submit practice and graded assignments, including programming ones, so that I can reinforce what I am learning as well as add a new dimension to exam preparation.
- As a student, I want course-specific chat rooms, so that I can interact with my peers and the course faculty for better learning and collaboration.
- As a student, I want an announcement/notification system, so that I don't miss any assignment deadlines and Instructor/TA live sessions.
- As a student, I want to keep track of my progress/milestones so that I can stay motivated and plan accordingly.
- As a student, I want to access all the PYQs of each course, so that I can get an idea regarding question patterns and distribution of marks for exams.
- As a student, I want a feedback and review system so that I can make sure that the quality of content being provided is improved whenever students find it necessary.
- As a student, I want a note-making space so that I can ensure not missing anything from the lecture which in turn shall provide a condensed record for efficient study and revision.

- As a student, I want the AI assistant to guide me through concepts when I am stuck, so that I can improve my understanding of the course material.
- As a student, I want the AI assistant to resolve my queries regarding various endpoints of the platform so that I can navigate through it smoothly.
- As a student, I want to interact with the AI assistant so that I can refine and improve my study habits.
- As a student, I want to be able to download important documents so that I can have local access to important material when needed.

2.2 Professors

- As a professor, I want to manage lectures and assignments so that I can provide structured learning for students.
- As a professor, I want to monitor student performance and course statistics so that I can identify concepts that need additional resources or focus.
- As a professor, I want to collaborate with instructors so that the workload is manageable.
- As a professor, I want access to a course-specific chat room so that I can engage with students and address their queries efficiently.
- As a professor, I want to stay connected to the management so that I can keep the platform aligned with institutional goals.
- As a professor, I want AI as my assistant so that I can streamline repetitive tasks, analyze student performance and provide additional resources or support for challenging concepts.

- As a professor, I want a separate chat room for course faculty (professors, instructors, and TAs) so that we can discuss course-related matters privately.

2.3 Instructors

- As an instructor, I want to assist professors in managing content so that I can help students have a better learning experience.
- As an instructor, I want to communicate with teaching assistants so that I can help manage student queries.
- As an instructor, I want to use the AI assistant so that I can take care of routine tasks, keep an eye on how students are doing, and offer extra help or materials when they're struggling with tricky ideas.
- As an instructor, I want to use chat rooms so that I can facilitate discussions and collaborations among students.
- As an instructor, I want to access student statistics/performance metrics so that I can identify areas where students need help.
- As an instructor, I want a separate chat room so that I can collaborate with professors and TAs privately.
- As an instructor, I want to set up announcements so that I can notify all students of deadlines and live sessions.

2.4 Teaching Assistants (TAs)

- As a teaching assistant, I want access to chatrooms so that I can assist students with their queries and support instructors in managing discussions.

- As a teaching assistant, I want to communicate with instructors and professors on a separate chat room so that I can collaborate effectively.

2.5 Admins

- As an admin, I want CRUD (Create, Read, Update, Delete) access so that I can manage all platform resources effectively.
- As an admin, I want to manage user roles and permissions so that I can ensure that the platform remains secure.
- As an admin, I want to manage chat rooms for courses so that I can ensure smooth communication and collaboration among users.

2.6 Degree Managers

- As a degree manager, I want CRUD (Create, Read, Update, Delete) access so that I can manage all platform resources effectively.
- As a degree manager, I want to oversee the system and make corrections so that I can ensure that the platform operates smoothly.
- As a degree manager, I want a separate forum for collaboration with admins and faculty so that I can ensure that the communication is streamlined.

2.7 Developers

- As a developer, I want to update and manage the codebase so that I can implement new features.
- As a developer, I want to resolve bugs quickly so that I can maintain the platform's stability.

2.8 Management

- As a manager, I want to oversee platform performance so that I can ensure that it aligns with institutional goals.
- As a manager, I want to make strategic decisions for platform enhancements so that I can add to the potential scalability.

3. Team Details

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